

Artificial Intelligence & Machine Learning — Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Dataset: Breast Cancer Dataset (scikit-learn) **Problem Type:** Binary Classification

1. Objective

The objective of Task 4 is to move from regression to binary classification and evaluate classification models using metrics beyond accuracy. The workflow includes Logistic Regression, confusion-matrix analysis, precision, recall, F1-score, ROC-AUC, class-imbalance handling, and comparison with a Decision Tree classifier.

2. Dataset and Preprocessing

The scikit-learn Breast Cancer dataset was used. It contains 569 observations, 30 numerical features, and two target classes: malignant (0) and benign (1). The data was divided using an 80/20 stratified train-test split with `random_state=42`. `StandardScaler` was fitted only on the training data and then applied to the training and test sets.

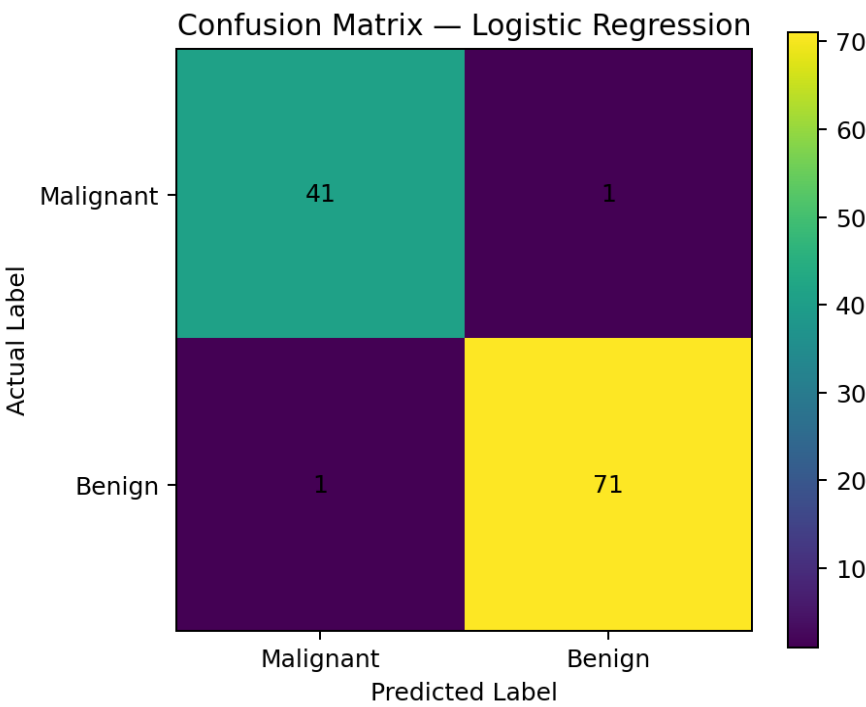
3. Models Implemented

Logistic Regression: used as the baseline classifier with `max_iter=1000`.
Balanced Logistic Regression: uses `class_weight='balanced'` to give additional consideration to the class distribution.
Decision Tree: used as a second classifier for comparison and trained on the unscaled features.

4. Evaluation Metrics

Accuracy is the proportion of all predictions that are correct. **Precision** indicates how reliable positive predictions are. **Recall** measures how many actual positive cases are detected. **F1-score** combines precision and recall and is useful when both types of error matter. **ROC-AUC** measures discrimination across classification thresholds. The confusion matrix provides the underlying counts of correct and incorrect predictions.

Accuracy should therefore not be considered in isolation when class proportions or error costs make precision and recall important. This is particularly relevant to the task's emphasis on medical-diagnosis examples, where missed cases can have different consequences from false alarms.



5. Confusion Matrix and Baseline Performance

The Logistic Regression confusion matrix on the test set was:

[[41, 1], [1, 71]].

The matrix separates predictions into true/false outcomes and makes it possible to see the types of classification errors rather than only the overall accuracy.

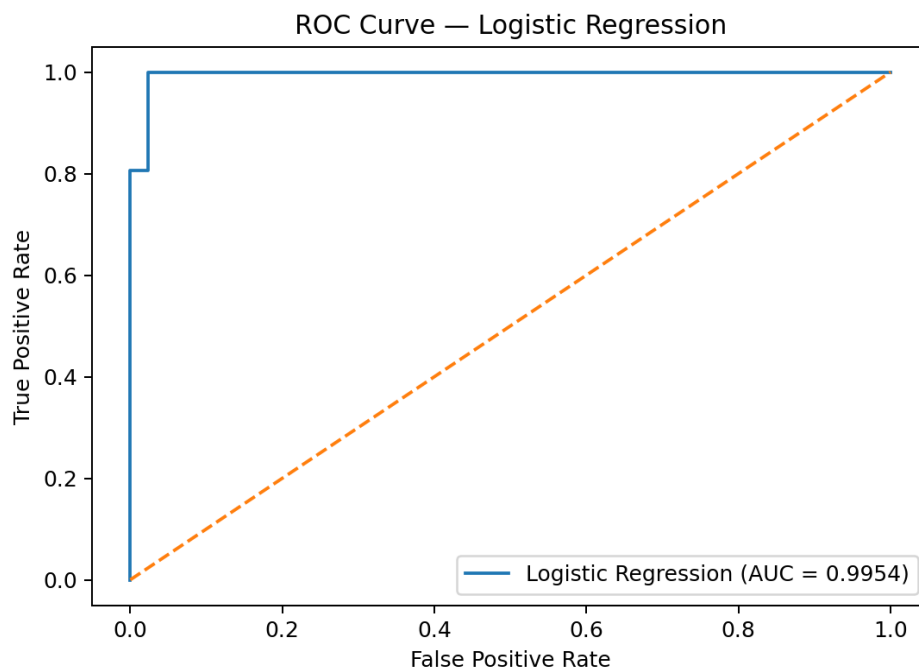
6. Model Comparison Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.982456	0.986111	0.986111	0.986111	0.995370
Balanced Logistic Regression	0.956140	0.985507	0.944444	0.964539	0.995370
Decision Tree	0.912281	0.955882	0.902778	0.928571	0.915675

The supplied model metrics show Logistic Regression with accuracy 0.982456, precision 0.986111, recall 0.986111, and F1-score 0.986111. Balanced Logistic Regression has lower accuracy (0.956140) and recall (0.944444), while maintaining high precision (0.985507). The Decision Tree records lower performance on the supplied accuracy, precision, recall, and F1-score values.

7. ROC Curve and AUC

Using the same train-test split and model configuration, the calculated ROC-AUC values are 0.995370 for Logistic Regression, 0.995370 for Balanced Logistic Regression, and 0.915675 for the Decision Tree. ROC-AUC complements the point metrics because it evaluates ranking/discrimination over different probability thresholds.



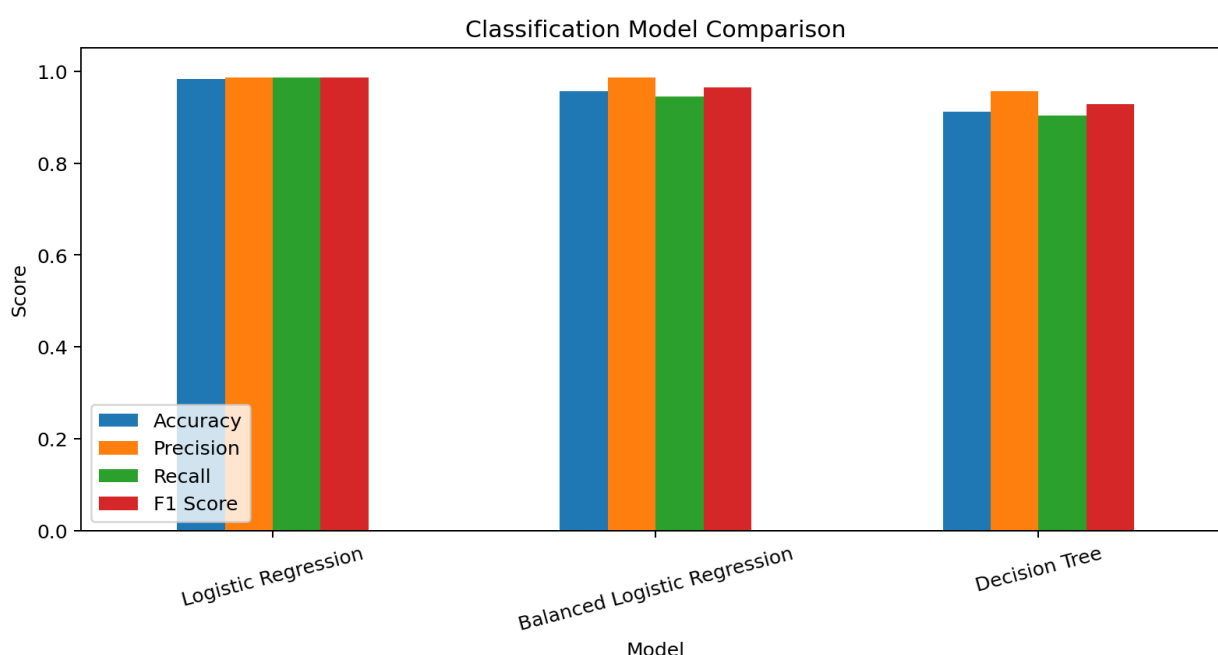
8. Handling Class Imbalance

Class imbalance occurs when one target class has substantially more observations than another. A model can sometimes achieve high accuracy while performing poorly on the less represented class. The task therefore asks for an imbalance-handling experiment using `class_weight='balanced'` in Logistic Regression. This changes the training objective so that errors associated with the classes receive weights based on their frequencies.

In this experiment, the balanced Logistic Regression model produced the supplied accuracy of 0.956140, precision of 0.985507, recall of 0.944444, and F1-score of 0.964539. These values should be interpreted together rather than using accuracy alone.

9. Logistic Regression vs Decision Tree

The comparison demonstrates that different classifiers can produce different error profiles. Logistic Regression is a linear probabilistic classifier and, in this experiment, achieved higher supplied accuracy, precision, recall and F1-score than the Decision Tree. A Decision Tree is easier to express as a sequence of decision rules, but unrestricted trees can be sensitive to training data and can overfit. The comparison should therefore consider both measured performance and model behavior.



10. Final Model Decision

Based on the supplied test-set metrics, Logistic Regression is the model selected for this Task 4 implementation. It has the highest supplied accuracy, precision, recall, and F1-score among the three reported models, while also providing a very high ROC-AUC under the same evaluation setup. The selection is therefore based on the complete metric profile rather than accuracy alone.

Note: The numerical Accuracy, Precision, Recall and F1-score values in this report are the values supplied from the user's completed notebook. ROC-AUC values were calculated from the same scikit-learn dataset, split, preprocessing and model configurations.